LIBRARY BOOK ISSUE COMMAND CENTRE

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR in

BACHELOR OF TECHNOLOGY
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, TamilNadu – 602 105

OCTOBER 2026



RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "LIBRARY BOOK ISSUE COMMAND CENTRE" is the bonafide work of [DIVYADHARSHINI.K] ([241501053]), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester [III], under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, [DIVYADHARSHINI.K] ([241501053]),hereby declare that the project report titled "LIBRARY BOOK ISSUE COMMANDCENTRE", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports and dashboard — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: [19] October 2026

SIGNATURE OF THE STUDENT
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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman [Name], Chairperson [Name] and Vice-Chairman [Name] for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, [Name], Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank [Name], Head of the Department, Department of [Department], for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, [Name], [Designation], for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for the Zoho Creator platform and its learning resources, and I thank my family and friends for their unwavering support.

DIVYADHARSHINI.K

ABSTRACT

Libraries lend books to many students every day. Records of student details, book copies, issues, returns and fines are often kept in registers, spreadsheets or separate files. When information is scattered in this way, the number of copies available is hard to confirm, the condition of returned books is not recorded consistently, and finding books that are damaged or missing requires a manual search of many records.

Library Book Issue Command Centre is a library circulation management application developed on the Zoho Creator low-code platform to bring these activities into one place. The application is organised into five navigation sections: Dashboard, Book Return, Issues, Book Inventories and Student Profiles. The Student Profile form captures the student name, student ID, email, phone number, class grade, department, enrollment date, status and address. The Book Inventory form captures the book name, ISBN, author, publisher, publication year, genre and total copies. Each book return is recorded with a return reference (for example RET-2026-005), return date, return condition (Good, MinorDamage, SevereDamage or Missing), the name of the person returning the book and notes. List reports display the stored records, a board view groups book inventories, and a dashboard summarises the total fine amount, total copies, number of book returns, number of issues, the distribution of students by status and the count of returns in each return condition.

The application stores, groups and counts records; it does not use a trained artificial intelligence or machine learning model, and no external integrations are used. Test cases covering data entry, reports and dashboard totals are listed in Chapter 6; those that can be verified from the application output are recorded as passed. The project demonstrates how the forms, reports and dashboard features of Zoho Creator reduce the amount of conventional code needed for a practical library record-keeping application, while showing that correct data modelling, validation and testing remain essential.

Keywords: Low-Code Development; Zoho Creator; Library Management; Book Issue and Return; Inventory Management; Dashboard Analytics

PROJECT AT A GLANCE

Attribute	*Details*
Project Title	LIBRARY BOOK ISSUE COMMAND CENTRE
Application Domain	Education – Library Book Issue and Return Management
Student Name & Register No.	[DIVYADHARSHINI.K]– [241501053]
Zoho Creator Edition	Zoho Creator Trial Edition (the trial banner showed 15 days remaining on 21 October 2026)
Application Name (Link Name)	Library Book Issue Command Centre (link name: library-book-issue-command-centre)
Published / Live URL	https://creatorapp.zoho.in/divyadharshinik53/library-book-issue-command-centre
Demo Credentials (per role)	Librarian / Application Owner: [Zoho account e-mail of the application owner]
Number of Forms / Subforms	4 forms (Student Profile, Book Inventory, Issue, Book Return) / [number of subforms, 0 if none] subforms
Number of Reports (by type)	List: [4 – All Student Profiles, All Book Inventories, Book Return Report, Issues report] Kanban / Board: 1 (Book Inventories Board) Chart: 1 (Distribution by Status pie chart on the Dashboard) Others: Return Conditions and Statuses views [state report type]
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	[Form / Schedule / Approval / Blueprint / Batch – enter counts]
Deluge Functions / Custom Actions	[Enter the Deluge scripts used, or “None”]
Integrations	[None / list integrations]
Roles & Profiles Defined	[Enter roles and profiles configured, or “Default administrator access only”]
Environments Used	Development environment of a single Zoho Creator workspace
Date of Production Deployment	[confirm] October 2026

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LIST OF ABBREVIATIONS

Abbreviation	*Expansion*
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
ISBN	International Standard Book Number
KPI	Key Performance Indicator
ML	Machine Learning
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Background

A library lends books to students throughout the academic year. Each time a book is issued or returned, the librarian must note which student has the book, how many copies of the title remain, whether the book came back in good condition, and whether any fine is due. Ranganathan's well-known principle that the time of the reader should be saved [6] applies equally to the time of the librarian: the quicker a transaction can be recorded and checked, the more time is left for service.

In many libraries these records are maintained in registers, spreadsheets or separate files for students, books, issues and returns. Student details, book copy counts, issue entries and return conditions are therefore stored in different places, and preparing even a simple summary requires repeated manual effort.

Library Book Issue Command Centre has been developed as an individual academic project for the course CB23F35 – Low-Code Application Development using Zoho Creator. It is a small-scale application intended for a librarian who wants to keep student profiles, the book inventory, book issues and book returns in one place, and to see a summary on a single dashboard. The main purpose of the project is to demonstrate how Zoho Creator forms, reports and dashboards can simplify library record keeping.

The sample records shown in this report — for example the books "The Great Gatsby" and "1984", and returns made by persons such as Eva Martinez and David Lee — are demonstration data. They do not represent real students or the holdings of an actual library.

1.2 Problem Statement

Librarians need a convenient way to maintain student profiles, keep track of the books and copies held, record which books are issued, and record the return of each book together with its condition. When these activities are performed manually or across separate files, record management requires additional effort, damaged or missing books are noticed late, and no quick overview of library activity is available. A simple, centralised application is therefore required. The specific pain points are:

- Student, book, issue and return information is scattered across registers, spreadsheets or separate files.
- The total number of copies held and the number of books issued are not visible at a glance.
- The condition of a returned book (good, damaged or missing) is not recorded in a consistent, searchable form.
- Fine amounts are noted by hand and are difficult to total.
- There is no quick visual summary of students by status or of returns by condition.

1.3 Objectives

- To create a centralised form for maintaining student profile information, including the status of each student.
- To maintain a book inventory with bibliographic details and the total number of copies.
- To record the issue of books to students.
- To record the return of each book with a return reference, return date, return condition, returned-by name and notes.
- To classify every return as Good, MinorDamage, SevereDamage or Missing so that problem returns can be identified.
- To provide reports, a board view and a dashboard for viewing library information. •

To demonstrate the use of Zoho Creator's low-code development features in an educational application.

1.4 Scope of the Project

Library Book Issue Command Centre focuses on basic library circulation management for a single library managed by a librarian. The scope includes:

- Student profile management (name, student ID, email, phone number, class grade, department, enrollment date, status and address).
- Book inventory management (book name, ISBN, author, publisher, publication year, genre and total copies).

- Recording of book issues and book returns, with the return condition and any fine amount.
- Reports, a board view and a dashboard presented through a simple, understandable user interface.

The application is accessed through a web browser on the Zoho Creator platform by the librarian, who is also the application owner in the current version.

Out of scope. The application does not use barcode or RFID scanning, does not process online fine payments, does not provide a student or customer portal, does not use a trained machine learning model and does not integrate with any external system. Features that are not implemented are documented as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

A low-code platform lets developers build applications mainly by configuring visual models of data, user interfaces and logic, with hand-written code needed only for specific requirements [1]. Zoho Creator provides a visual form builder, automatically generated reports, a page builder for dashboards and a hosted cloud environment, with the Deluge scripting language available when custom logic is required [2], [5].

The library application is well suited to this approach. Its data consist of structured records (students, books, issues and returns), and its main requirements are data-entry forms, record lists, a board view and a summary dashboard — all of which map directly onto standard Zoho Creator components. Building the same application conventionally would require a separately designed database, server-side code, user interface pages and hosting.

Using Zoho Creator therefore reduces the amount of conventional code to be written, removes the need to set up a server or database, and makes it easier to add or change fields later. A low-code platform does not remove the need for careful design: correct data modelling, validation of entered data, testing and access control still have to be planned and verified by the developer.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional approach, student details, the list of books and copy counts, book issue entries and return details are commonly maintained in separate registers or spreadsheets. The librarian writes or types the student and book details, records each issue by hand, notes the condition of a returned book and any fine in a separate place, and searches through the records to find pending, damaged or missing books before preparing any summary. Figure 2.1 shows this general as-is process. It describes the general limitations of manual record keeping and is not a claim about the exact process followed at any particular institution.

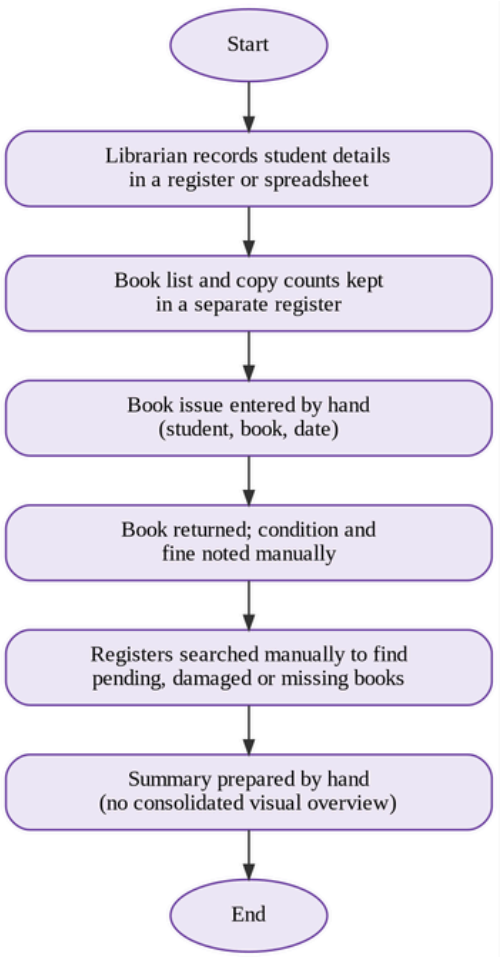


Figure 2.1: As-Is Process Flow of the Existing (Manual) System

2.2 Limitations of the Existing System

- ***Scattered information:*** student, book, issue and return records are kept in different places.
- ***Manual record keeping:*** every issue, return and fine is written or typed by hand. •

No combined view: it is difficult to see students, books and transactions together. •

Late detection of problem returns: damaged or missing books are found only when records are searched manually.

- ***Limited visual summaries:*** there is no ready overview of library activity.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a conventional (pro-code) implementation of the library application with its low-code implementation on Zoho Creator. The criteria are adapted from feature-based comparisons of low-code development platforms [3]. Development effort is described qualitatively because the time taken was not formally measured.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	***Traditional (Pro-Code) Development***	***Low-Code on Zoho Creator***
Development time	Longer: the database schema, server-side code, user interface pages and hosting must each be built and connected.	Shorter: forms, reports and the dashboard are configured visually; the actual effort was not formally measured.
Skills required	A programming language, a web framework, SQL and deployment skills.	Data modelling and Zoho Creator configuration; Deluge scripting only where custom logic is needed.
Hosting & infrastructure	A web server and database must be provisioned and maintained separately.	Hosted on the Zoho Creator cloud; no server set-up is required from the developer.
Mobile application	A responsive design or a separate mobile application must be developed.	Zoho Creator offers mobile access to applications; not tested for this application.
Security & access control	User authentication and authorisation must be coded and tested.	Uses Zoho account sign-in and platform permission settings.
Maintenance & change	Each change requires code modification, testing and redeployment.	Fields, forms and reports can be modified in the visual builder with little or no code.
Customisation limits	Fully flexible user interface and logic.	Limited to the components, layouts and scripting features offered by the platform.

Criterion	*Traditional (Pro-Code) Development*	*Low-Code on Zoho Creator*
Cost	Developer time plus separate hosting costs.	Depends on the Zoho Creator plan used; no separate hosting cost.

2.4 Literature Survey

The sources reviewed fall into two groups. The first group consists of studies of low-code development platforms [1], [3], [4] and the official Zoho Creator and Deluge documentation [2], [5], which inform how the application is built. The second is a classic principle of library science [6], which provides general background on why quick, accurate circulation records matter. Table 2.2 summarises these sources. The library-science source is background only: the application implements record keeping, reports and a dashboard, not the methods discussed in it.

Table 2.2: Summary of Literature Survey

S.No.	*Author(s) / Source & Year*	*Title / Product*	*Key Findings*	*Relevance to this Project*
1	A. C. Bock and U. Frank, 2021 [1]	Low-Code Platform	Describes low-code platforms as environments in which applications are built mainly by configuring models of data, user interfaces and processes, reducing hand-written code while tying the application to the concepts of the platform.	Explains why a form-and-report based application such as this one suits a low-code platform, and the trade-offs involved.
2	A. Sahay et al., 2020 [3]	Supporting the Understanding and Comparison of Low-Code Development Platforms	Analyses a selection of low-code development platforms and proposes a set of features for understanding and comparing them.	Provides the basis for the comparison criteria used in Table 2.1.
3	D. Di Ruscio et al., 2022 [4]	Low-Code Development and Model-Driven Engineering: Two Sides of the Same Coin?	Discusses the relationship between low-code development and model-driven engineering, and the opportunities and challenges they share.	Helps explain that the forms act as models from which Zoho Creator generates the data store and user interface.
4	S. R. Ranganathan, 1931 [6]	The Five Laws of Library Science	Sets out five principles of library service, including that the time of the reader should be saved.	Background for providing a fast, single-place record of issues and returns.
5	Zoho Corporation [2], [5]	Zoho Creator Help and Deluge Help	Official documentation of Zoho Creator forms, fields, reports, pages, workflows and the Deluge language.	Primary technical reference used to design and build the application.

2.5 Emerging Trends in Low-Code Development

AI-assisted application building. Low-code platforms increasingly offer AI-assisted features, such as generating application structures or scripts from natural-language descriptions. [State here whether any AI-assisted feature was or was not used to build this application.]

Citizen development. Low-code tools allow people with domain knowledge but limited programming experience to build simple applications. The structure of this application (students, books, issues, returns) mirrors how a librarian already thinks about circulation, so it can be understood and modified without deep programming knowledge.

Convergence with model-driven engineering. Low-code development is closely related to model-driven engineering, in which models are the main development artefacts [4]. Here, the form definitions act as the model from which the platform generates storage and user interface.

2.6 Proposed System

The proposed system, Library Book Issue Command Centre, is a centralised Zoho Creator application that contains forms to capture student, book, issue and return information, reports and a board view to display the saved records, and a dashboard to summarise the available information. Every return is classified by a Return Condition (Good, MinorDamage, SevereDamage or Missing), and the dashboard shows how many returns fall under each condition. The classification is chosen by the librarian when the return is recorded; it indicates the state of the returned book and does not by itself establish who is responsible for damage or loss.

The proposed system addresses the limitations listed in Section 2.2 as follows: scattered information is replaced by one application with related forms; manual entries are replaced by structured forms; the reports and board view provide a combined view of records; the Return Condition field makes damaged and missing books easy to locate; and the dashboard provides a visual summary.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	***Description***	***Key Responsibilities in the Application***
Librarian (Primary User)	The person who uses the application to manage library circulation.	Add and update student profiles and book inventory; record issues and returns; view reports, the board view and the dashboard.
Application Developer / Administrator	The student developer who builds and owns the Zoho Creator application.	Design and maintain forms, fields, reports and the dashboard; manage application settings.
Students (Indirect Stakeholders)	Borrowers whose profiles, issues and returns are recorded in the application.	No direct access in the current version; they benefit from faster and more accurate circulation records.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	***Requirement***	***Role(s)***	***Priority***
FR-01	The system shall allow the librarian to add, view and update student profiles with Student Name, Student ID, Email, Phone Number, Class Grade, Department, Enrollment Date, Status and Address.	Librarian	High
FR-02	The system shall allow the librarian to add, view and update book inventory records with Book Name, ISBN, Author, Publisher, Publication Year, Genre and Total Copies.	Librarian	High
FR-03	The system shall allow the librarian to record the issue of a book to a student.	Librarian	High
FR-04	The system shall allow the librarian to record a book return with Return Reference, Return Date, Return Condition, Returned By and Notes.	Librarian	High
FR-05	The system shall classify each return as Good, MinorDamage, SevereDamage or Missing.	Li br ar ian	High
FR-06	The system shall record fine amounts and display their total.	Li br ar ian	Medium
FR-07	The system shall provide reports listing the records of each module.	Li br ar ian	Medium
FR-08	The system shall provide a board view of book inventories.	Li br ar ian	Low
FR-09	The system shall record the status of each student (Active, Graduated or Inactive) and show the distribution of students by status.	Li br ar ian	Medium

FR ID	Requirement	Role(s)	Priority
FR-10	The system shall display a dashboard with total fine amounts, total copies, the number of book returns, the number of issues, the distribution of students by status, the count of returns by condition and the Book Return Report.	Librarian	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports shall open within an acceptable time for a library-sized data set (not formally measured in this project).
NFR-02	Secur ity	Student data shall be accessible only to authorised Zoho account users of the application.
NFR-03	Usability	The interface shall use clear labels, a simple navigation menu and a minimal number of fields per form.
NFR-04	Availability	The application shall be available wherever the Zoho Creator service and an internet connection are available.
NFR-05	Maintainability	Fields, forms and reports shall be modifiable through the Zoho Creator builder without rewriting the application.
NFR-06	Data Integrity	Each return shall carry a Return Condition, and dashboard totals shall agree with the records in the reports.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial Edition)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment) – [state where Deluge is used, or “not used”]
Browser	A modern web browser such as Google Chrome or Microsoft Edge (latest version)
API testing tool	Not required (no API integration in the current version) [confirm]
Other Zoho services	[None used / list]
Mobile	Not used in the current version
Client hardware	Desktop computer or laptop with keyboard, mouse and a reliable internet connection; no specialised hardware is required
Documentation	Microsoft Word or a compatible document editor for the project report

3.5 Use Case Model

Figure 3.1 shows the use case diagram of the application. The librarian is the primary actor who manages records and views reports; the application developer/administrator configures the application. Students are indirect stakeholders and do not interact with the application. The key use case is described in Table 3.5.

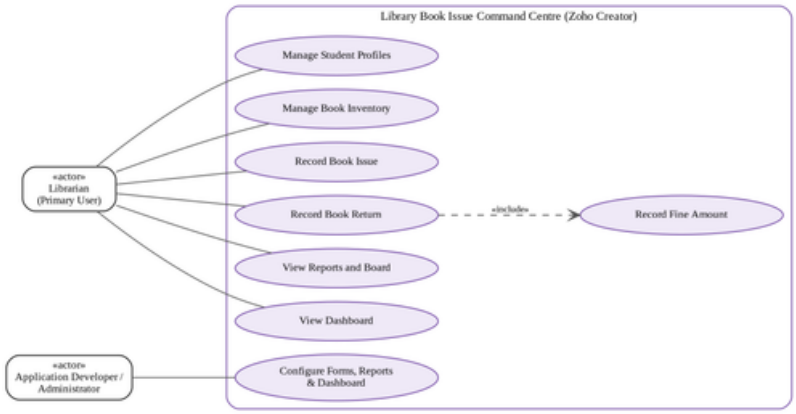


Figure 3.1: Use Case Diagram of the Library Book Issue Command Centre

Table 3.5: Use Case Description – Record Book Return and Review Return Condition

Item	***Description***
Use Case ID	UC-01
Use Case Name	Record Book Return and Review Return Condition
Actor(s)	Librarian
Pre-condition	The student profile and book inventory records exist and the book has been issued.
Main Flow	1. The librarian opens the Book Return form. 2. The librarian enters the return details: Return Date, Returned By and Notes. 3. The librarian selects the Return Condition (Good, MinorDamage, SevereDamage or Missing) and submits the form. 4. The application saves the record with a return reference of the form RET-2026-nnn. 5. The record appears in the Book Return Report and the dashboard counts are updated.
Alternate Flow	If a mandatory field is left empty, the application displays a prompt and the record is not saved.
Post-condition	A valid return record is stored with its return condition.
Related FR	FR-04, FR-05, FR-07, FR-10

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

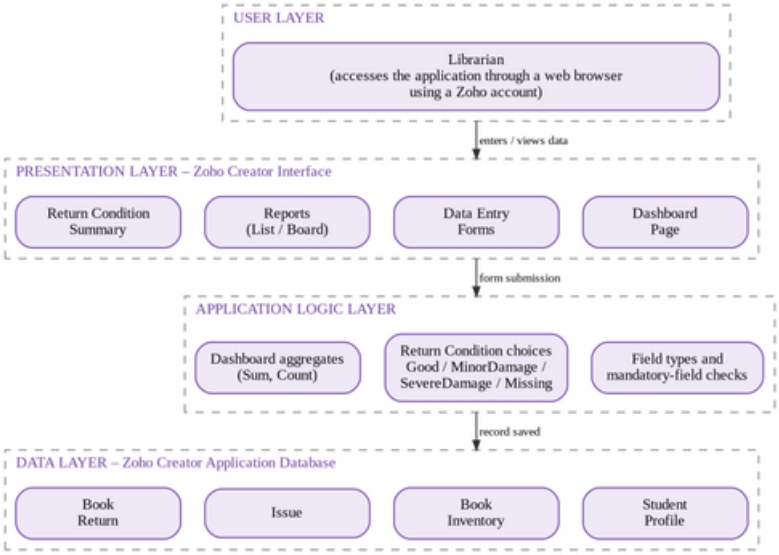


Figure 4.1: System Architecture of the Library Book Issue Command Centre

Figure 4.1 shows the layered architecture of the application. The flow of information is: User → Zoho Creator Interface → Forms and Reports → Application Data → Dashboard.

User layer. The librarian accesses the application through a web browser after signing in with a Zoho account.

Presentation layer. The Zoho Creator interface presents the Dashboard page, the data-entry forms, the list and board reports and the return condition summary. These are generated by the platform from the configured forms, reports and page.

Application logic layer. When a form is submitted, the entered values are checked against the field types and mandatory fields, the Return Condition is stored as one of four choices, and the dashboard aggregates (sum and count) are computed from the stored records. [Add any Deluge or workflow logic you implemented.]

Data layer. Submitted records are stored in the built-in database of the Zoho Creator application, one record set per form (Student Profile, Book Inventory, Issue and Book Return). The reports and the dashboard read these stored records. No external database or integration is used.

4.2 Data Model

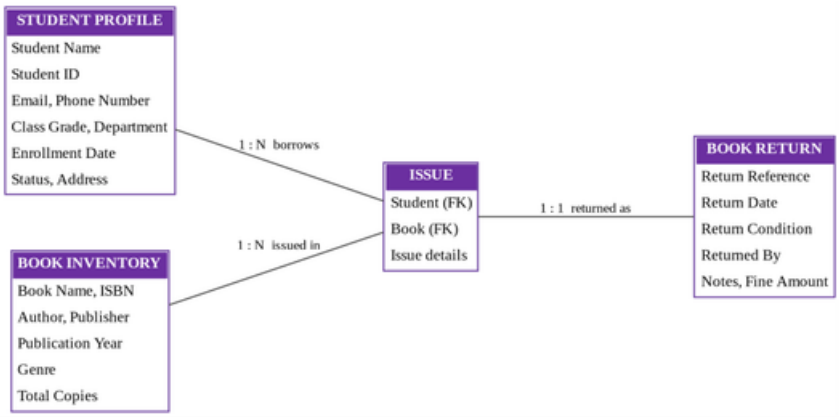


Figure 4.2: Entity–Relationship Diagram of the Library Book Issue Command Centre (Conceptual)

The application has four entities, each implemented as a Zoho Creator form. A student can have many issues, a book can be issued many times, and each issue is closed by a return. In Zoho Creator, such relationships are realised through lookup fields. The relationships in Figure 4.2 and Table 4.1 represent the conceptual relational model of the application. [Confirm the lookup fields you actually used in the Issue and Book Return forms and correct Table 4.1 if necessary.]

Table 4.1: Entities and Relationships

Entity (Form)	*Key Fields*	*Related To*	*Relationship Type*	*Implemented Using*
Student Profile	Student Name, Student ID, Email, Status	Issue	One-to-Many	Lookup to Student Profile in the Issue form
Book Inventory	Book Name, ISBN, Author, Total Copies	Issue	One-to-Many	Lookup to Book Inventory in the Issue form
Issue	Issue details	Book Return	One-to-One / One-to-Many	Lookup to Issue in the Book Return form
Book Return	Return Reference, Return Date, Return Condition, Returned By, Notes	Issue	Many-to-One	Lookup field (see above)

4.3 Form Design

Tables 4.2 to 4.5 give the field specification of the four forms. The link names shown are the default link names that Zoho Creator generates from the field labels. Field types and choices are taken from the application screens (Figures 5.2 to 5.4).

Table 4.2: Field Specification – Student Profile Form

Field Label	*Link Name*	*Field Type*	*Validation / Default*	*Remarks*
Student Name	Student_Name	Name	First Name sub-field	Name of the student
Student ID	Student_ID	Single Line	—	Identifier of the student
Email	Email	Email	Valid e-mail format	Contact e-mail
Phone Number	Phone_Number	Phone	Default country code +91 (India)	Contact number
Class Grade	Class_Grade	Single Line	—	Class or grade of the student
Department	Department	Single Line	—	Department of the student
Enrollment Date	Enrollment_Date	Date	Format dd-MMM-yyyy	Date of enrollment
St at us	Status	Drop Down	Choices: Active, Graduated, Inactive	Drives the Distribution by Status chart
Address	Address	Address	Address Line 1 and further lines	Postal address

Table 4.3: Field Specification – Book Inventory Form

Field Label	*Link Name*	*Field Type*	*Validation / Default*	*Remarks*
Book Name	Book_Name	Single Line	—	Title of the book
ISBN	ISBN	Single Line	—	e.g. 978-0743273565
Author	Author	Single Line	—	Author of the book
Publisher	Publisher	Single Line	—	Publisher of the book
Publication Year	Publication_Year	Number	—	Year of publication
Genre	Genre	Drop Down	Fixed set of genre choices	Grouping of books by genre
Total Copies	Total_Copies	Number	—	Number of copies held; summed on the dashboard
[Other fields]	[…]	[…]	[…]	[Add any further fields of the form]

Table 4.4: Field Specification – Issue Form

Field Label	*Link Name*	*Field Type*	*Validation / Default*	*Remarks*
[Field 1]	[…]	[…]	[…]	[…]
[Field 2]	[…]	[…]	[…]	[…]
[Field 3]	[…]	[…]	[…]	[…]

Table 4.5: Field Specification – Book Return Form

Field Label	*Link Name*	*Field Type*	*Validation / Default*	*Remarks*
Return Reference	Return_Reference	Single Line / Auto Number [confirm]	Format RET-2026-nnn	Unique reference of each return
Return Date	Return_Date	Date	Format dd-MMM-yyyy	Date on which the book was returned
Return Condition	Return_Condition	Drop Down	Choices: Good, MinorDamage, SevereDamage, Missing	Colour-coded in the report
Returned By	Returned_By	Single Line / Lookup [confirm]	—	Person who returned the book
Notes	Notes	Multi Line	—	Free-text remarks
Fine Amount	Fine_Amount	Currency / Decimal [confirm]	—	Summed as Total Fine Amounts on the dashboard
[Issue / Book lookup]	[…]	Lookup	[…]	Links the return to its issue or book

4.4 Process and Workflow Design

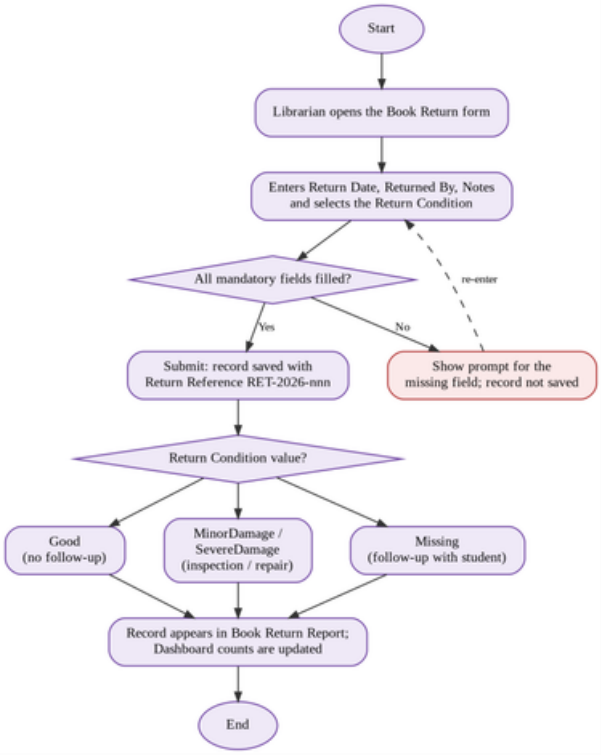


Figure 4.3: Process Flow for Recording a Book Return and Classifying Its Condition

Figure 4.3 shows the process followed when a book is returned. The librarian enters the return details and selects the Return Condition; if a mandatory field is missing, the form

prompts for it and the record is not saved. The saved record receives a return reference and becomes visible in the Book Return Report and the dashboard. The overall process is: add student profiles → add books to the inventory → issue books → record returns with their condition → view reports and dashboard summaries → follow up on damaged or missing books.

4.5 User Interface and Navigation Design

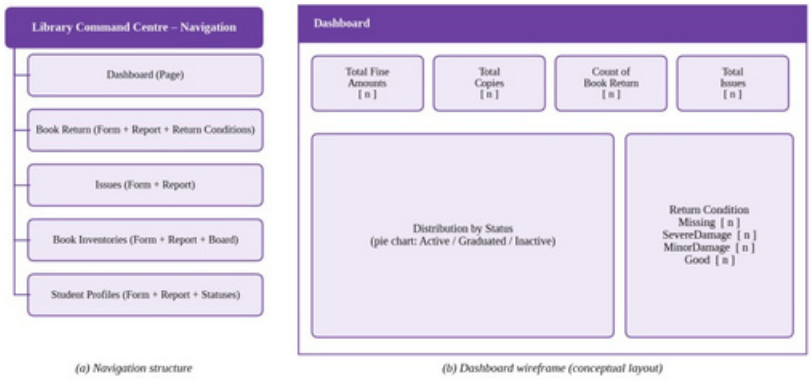


Figure 4.4: Navigation Structure and Wireframe of the Dashboard Page

The application is organised into five navigation sections that match its modules (Figure 4.4a): Dashboard, Book Return (Book Return form, Book Return Report and Return Conditions), Issues, Book Inventories (Book Inventory form, All Book Inventories and Book Inventories Board) and Student Profiles (Student Profile form, All Student Profiles and Statuses). The Dashboard is the first section so that the librarian sees the summary on opening the application. The wireframe in Figure 4.4b is the conceptual layout of the dashboard: four summary cards above a status distribution chart and a return condition summary. A dark navy side menu with green highlights is used throughout. It is a design sketch, not a screenshot of the implemented page.

4.6 Access Control Design

Table 4.6 defines the intended access design using A = Add, E = Edit, V = View, D = Delete and — = No access. The administrator (application owner) has full access; a librarian role, if configured, would add, edit and view records, with deletion reserved for the administrator to prevent accidental loss of data. Students have no access because the application does not provide a student portal. [State whether this role was configured; if not, say that the librarian uses the application through the owner account.]

Table 4.6: Role–Permission Matrix

Component	***Administrator (Application Owner)***	***Librarian***	***Student***
Form: Student Profile	A E V D	A E V	—
Form: Book Inventory	A E V D	A E V	—
Form: Issue	A E V D	A E V	—
Form: Book Return	A E V D	A E V	—
Reports and Board	V	V	—
Page: Dashboard	V	V	—

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application "Library Book Issue Command Centre" was created as a new application in the developer's Zoho Creator account (hosted at creatorapp.zoho.in) using the visual Form Builder and Page Builder. The application is organised into five navigation sections that correspond to its modules: Dashboard, Book Return, Issues, Book Inventories and Student Profiles (Figure 4.4 and Figure 5.1). A dark navy menu with green highlights and readable labels keeps the interface simple for a librarian.

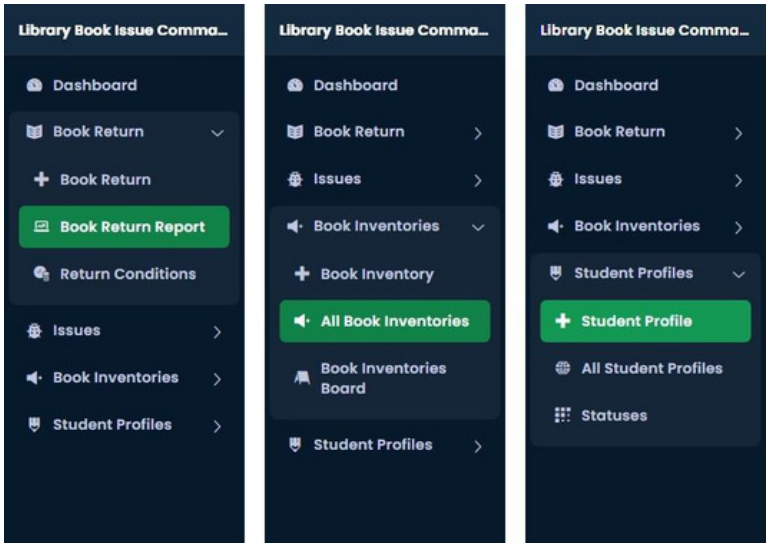


Figure 5.1: Application Navigation Sections (Book Return, Book Inventories and Student Profiles Expanded)

5.2 Forms and Data Capture

The application uses four forms, designed as specified in Tables 4.2 to 4.5. Each submitted form creates a record in the application database, which then appears in the corresponding report.

Student Profile form. Captures Student Name, Student ID, Email, Phone Number (with the +91 country code for India), Class Grade, Department, Enrollment Date (dd- MMM-yyyy), Status and Address, as shown in Figure 5.2. The Status drop-down classifies each student as Active, Graduated or Inactive.

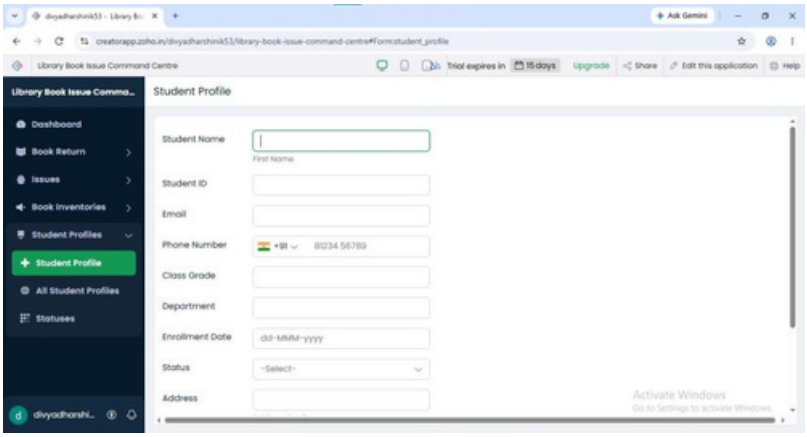


Figure 5.2: Student Profile Form in Live Mode

Book Inventory form. Stores the bibliographic details of each title — Book Name, ISBN, Author, Publisher, Publication Year and Genre — together with the number of copies held.

Book Return form. Records each return with a Return Reference (for example RET-2026-005), the Return Date, the Return Condition (Good, MinorDamage, SevereDamage or Missing), the person who returned the book (Returned By) and Notes.

5.3 Reports

Zoho Creator creates a list report for each form, through which records can be viewed, searched, edited and deleted [2]. The application uses these reports for its four modules and adds a board view for book inventories. Table 5.1 lists the reports.

All Book Inventories. This report lists each book with its Book Name, ISBN, Author, Publisher, Publication Year, Genre and Total Copies. The report footer shows “Showing 5 of 5”, confirming the number of stored records (Figure 5.3).

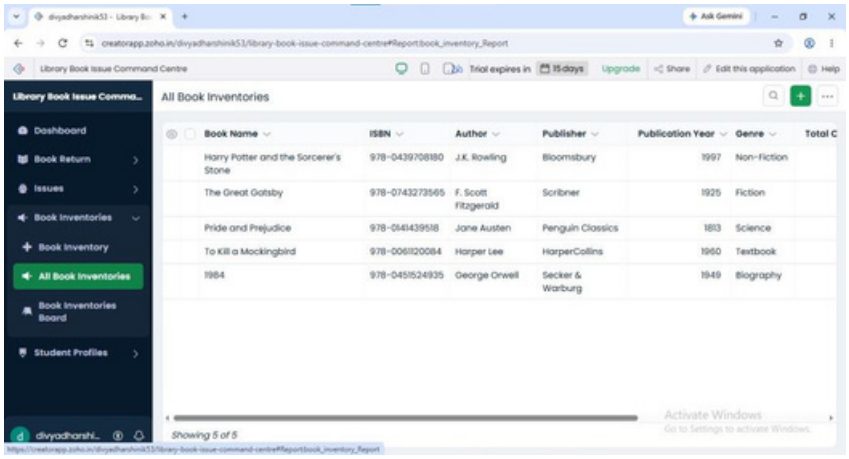


Figure 5.3: All Book Inventories Report

Book Return Report. This report lists each return with its Return Reference, Return Date, Return Condition, Returned By and Notes. The Return Condition cells are colour-coded so that MinorDamage and Missing returns stand out at a glance (Figure 5.4). The report footer shows “Showing 5 of 5”.

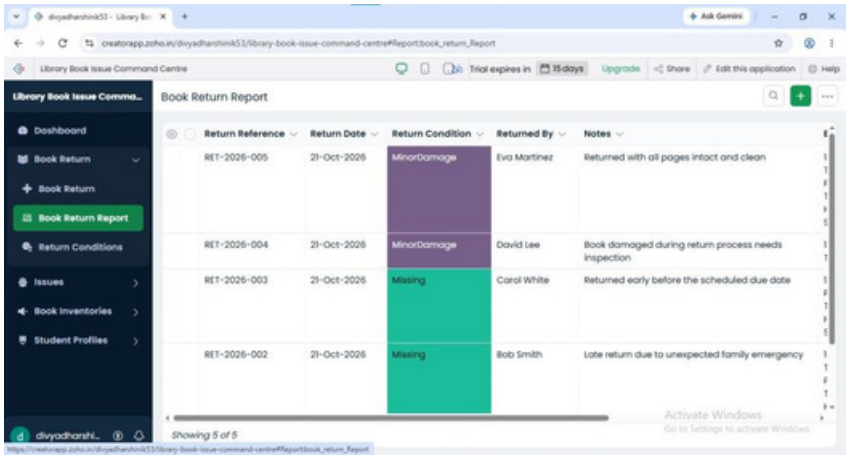


Figure 5.4: Book Return Report

Table 5.1: Reports Implemented

Report Name	***Type***	***Source Form***	***Filter / Grouping / Sorting***	***Roles***
All Student Profiles	List	Student Profile	[sorting / grouping used]	Librarian
All Book Inventories	List	Book Inventory	Sortable columns; search and add options	Librarian
Book Inventories Board	Board (Kanban)	Book Inventory	Cards grouped by a field [state field]	Librarian
Book Return Report	List	Book Return	Colour-coded Return Condition; newest return first	Librarian
Return Conditions	[t ype]	Book Return	Grouped by Return Condition [confirm]	Librarian
Statuses	[t ype]	Student Profile	Grouped by Status [confirm]	Librarian
Issues report	List	Issue	[…]	Librarian

5.4 Pages and Dashboards

The Dashboard page gives the librarian an overview of the application's records (Figure 5.5). It is built with the Zoho Creator page builder and contains:

- four summary cards — Total Fine Amounts ($ 0.00), Total Copies (870.00), Count of Book Return (5) and Total Issues (3);
- a pie chart, "Distribution by Status", showing students grouped as Active, Graduated and Inactive;
- a "Return Condition" panel showing the number of returns that are Missing (3), SevereDamage (0), MinorDamage (2) and Good (0); and
- the Book Return Report embedded below the charts.

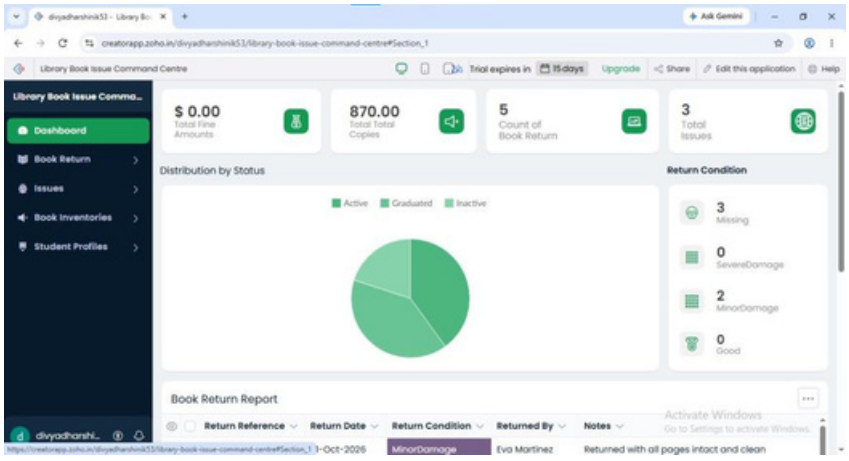


Figure 5.5: Dashboard Page

5.5 Deluge Scripting

Deluge is the scripting language used in Zoho Creator to add custom logic to forms, workflows and functions [5]. [Describe the Deluge scripts you wrote (for example on form events), or state that no Deluge script was required and the application was built with standard configuration only. If scripts exist, list them in Table 5.2, show the code as Code 5.1, 5.2 … and copy the full code to Appendix C.]

Table 5.2: Deluge Scripts Implemented

Script ID	*Location / Trigger*	*Purpose*	*Deluge Concepts Used*
[DS-01]	[Form → event]	[…]	[…]

Script ID	*Location / Trigger*	*Purpose*	*Deluge Concepts Used*
[DS-02]	[Form → event]	[…]	[…]

5.6 Workflows and Automation

[Describe the form, schedule, approval, Blueprint or batch workflows you configured, or state that none were used. Table 5.3 lists the workflow types and their status.]

Table 5.3: Workflows Implemented

Workflow Name	*Type*	*Trigger / Event*	*Condition*	*Action(s)*	*Script Ref.*
[…]	[Form Workflow]	[…]	[…]	[…]	[…]
Approval Process	Not implemented	—	—	Notrequired for the current scope	—
Blueprint	Not implemented	—	—	Notrequired for the current scope	—
Scheduled Workflow	Not implemented [confirm]	—	—	Proposed for overdue reminders (Section 9.2)	—

5.7 Integrations

No integration with other Zoho services or external systems is used in the current version [confirm]. The application does not call any REST API and does not expose a webhook. All data are entered and stored within the Zoho Creator application itself. Possible future integrations, such as e-mail reminders for overdue books, are listed in Section 9.2.

5.8 User Management and Role-Based Access

[Describe the users, roles and permission profiles configured in the application, or state that the application is used through the owner's Zoho account and no custom roles or portal are configured.] Implementing the role–permission matrix of Table 4.6 is planned as a future enhancement if it has not been configured. Until then, access to student data depends on keeping the application private to the owner account.

5.9 AI and Analytics Features

The analytics in the application are descriptive summaries: the record totals, the Distribution by Status chart and the Return Condition counts on the Dashboard, and the colour-coded Return Condition column of the Book Return Report (Figures 5.4 and 5.5). These summaries help the librarian see the overall picture but depend entirely on the accuracy and completeness of the entered records. No machine learning model has been trained or used, and the application does not predict future behaviour. [State whether any Zia / AI feature of Zoho Creator was used.]

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom JavaScript widgets or the Zoho Creator REST API) has been used [confirm]. The requirements of the project — forms, reports, a board view and a dashboard — are covered by the standard low-code features of the platform, so adding pro-code components would increase complexity without adding value at this stage.

5.11 Mobile and Portal Access

The application has been designed and used through a web browser. Access through the Zoho Creator mobile app has not been tested, and no customer portal is configured, because the only intended user is the librarian.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

The testing strategy covers the following levels:

- ***Form testing:*** each form is tested with valid input, missing mandatory fields and invalid values such as a malformed e-mail address.
- ***Report testing:*** saved records are checked in the reports, and the record count shown in each report footer is compared with the number of records entered.
- ***Dashboard testing:*** dashboard totals are compared with the number of records and values in the underlying reports.
- ***User acceptance testing:*** at least two users try the application in the librarian role and give feedback (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	*FR Ref.*	*Test Description*	*Input / Steps*	*Expected Result*	*Actual Result*	*Status*
TC-01	FR-01	Open the Student Profile form	Open Student Profiles → Student Profile	Form shows Student Name, Student ID, Email, Phone Number, Class Grade, Department, Enrollment Date, Status and Address	All nine fields displayed (Figure 5.2)	Pass
TC-02	FR-01	Submit a student profile with valid information	Fill all fields with sample data and submit	Record saved and listed in All Student Profiles	[Execute and record]	[To be executed]
TC-03	FR-01	Submit a student profile with an invalid e-mail	Email: “abc@”	E-mail format error; record not saved	[Execute and record]	[To be executed]
TC-04	FR-02	View the book inventory report	Open All Book Inventories	Books listed with ISBN, Author, Publisher, Publication Year and Genre	Five books listed; footer “Showing 5 of 5” (Figure 5.3)	Pass
TC-05	FR-02	Add a new book	Open Book Inventory form, enter details and submit	Book saved and listed in the report	[Execute and record]	[To be executed]
TC-06	FR-04	View the Book Return Report	Open Book Return → Book Return Report	Returns listed with Return Reference, Return Date, Return Condition, Returned By and Notes	Five returns listed, RET-2026-005 to RET-2026-001; footer “Showing 5 of 5” (Figure 5.4)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-07	FR -0 5	Return Condition is shown for each return	Inspect the Return Condition column	Each return shows one of the four conditions, colour-coded	MinorDamage and Missing values shown in colour-coded cells (Figure 5.4)	Pass
TC-08	FR-10	Dashboard count of returns matches the report	Compare Count of Book Return with the Book Return Report	Both show the same number	Dashboard shows 5; report shows 5 records	Pass
TC-09	FR-10	Return Condition totals add up to the number of returns	Add the four Return Condition counts	Sum equals Count of Book Return	Missing 3 + SevereDamage 0 + MinorDamage 2 + Good 0 = 5	Pass
TC-10	FR-09	Distribution by Status chart	Open Dashboard	Pie chart shows Active, Graduated and Inactive segments	Pie chart with the three status segments displayed (Figure 5.5)	Pass
TC-11	FR-03	Total Issues matches the Issues report	Compare Total Issues with the Issues report	Both show the same number	Dashboard shows 3; [Issues report count]	[To be executed]
TC-12	FR-02	Total Copies matches the inventory	Add the Total Copies column of All Book Inventories	Sum equals Total Copies on the Dashboard (870)	[Record the sum]	[To be executed]
TC-13	FR-04	Submit a return with a mandatory field empty	Leave a mandatory field blank and submit	Mandatory-field prompt; record not saved	[Execute and record]	[To be executed]

Test cases TC-01, TC-04 and TC-06 to TC-10 were verified against the application screens shown in Chapter 5. The remaining test cases require data entry in the live application and are marked “To be executed” until their results are recorded.

6.3 Debugging

Faults in a Zoho Creator application can be located by (i) re-submitting the form with boundary and invalid values; (ii) comparing dashboard figures with the underlying report records; (iii) the error messages displayed by Zoho Creator on form submission; and (iv) “info” statements and the execution logs when Deluge scripts are used. [Describe the problems you actually met while building the application and how you found them.]

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
[BUG-01]	[…]	[…]	[…]	[Closed]
[BUG-02]	[…]	[…]	[…]	[Closed]

6.5 User Acceptance Testing

[Record the peers or users who tested the application in the librarian role, their feedback and the action taken, in Table 6.3.]

Table 6.3: User Acceptance Feedback

User	*Role Tested*	*Feedback*	*Action Taken*
[Reviewer 1]	Librarian	[…]	[…]
[Reviewer 2]	Librarian	[…]	[…]

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator supports separate Development, Stage and Production environments in its supported plans, allowing changes to be tested before they are published to users. The availability of these environments depends on the Zoho Creator edition being used. For this application, development, testing and verification were carried out in the Zoho Creator Development environment under the developer's account in October 2026, on a trial edition of the platform. Because the application is intended for a single librarian (the application owner), it is accessed directly within the authenticated Zoho Creator workspace without separate multi-stage environment promotion.

Table 7.1: Deployment Checklist

S.No.	*Checklist Item*	*Done (✓)*
1	All test cases executed in Development / Stage	[✓ after Table 6.1 is completed]
2	Sample/test data removed or isolated	✓ (the records shown are fictional demonstration data)
3	Roles, permissions and users configured in Production	[N/A (single-owner access) / ✓]
4	Connections authorised in Production	N/A (no connections) [confirm]
5	Schedules and workflows enabled	[✓ / N/A]
6	Application published and URL verified	✓ (verified in the owner workspace)
7	Version / release notes recorded	✓ (Version 1.0 – October 2026)

7.2 Performance Optimisation

The application stores a small, library-sized data set (five records in each of the sample reports), so performance was not a major concern and no measurements were taken. Table 7.2 therefore records the good practices followed in the design; before-and-after values are not reported because they were not measured.

Table 7.2: Performance Optimisation Results

Optimisation Applied	*Component*	*Before*	*After*	*Improvement*
Only the necessary fields in each form	All four forms	Not measured	Not measured	Simpler data entry
Summary values computed by the platform (Sum, Count) instead of manual totals	Dashboard cards	Not measured	Not measured	Consistent totals
Colour-coded Return Condition column	Book Return Report	Not measured	Not measured	Easier review of problem returns

7.3 Security Features

The security measures relevant to this application are basic and are listed in Table 7.3. Access to the application requires signing in with the appropriate Zoho account, and student information should not be exposed unnecessarily. Record-level permissions and custom roles are [not configured in the current version / configured as described in Section 5.8].

Table 7.3: Security Controls Implemented

Security Control	*Where Applied*	*Threat Mitigated*
Zoho account sign-in	Access to the whole application	Unauthorised access
Application kept private to the owner account (not published publicly) [confirm]	Application sharing settings	Unnecessary exposure of student information
Field types and mandatory fields	All forms	Invalid or incomplete data
~~Record-level / role-based permissions~~ [Not configured (future enhancement) / describe]		Accidental editing or deletion of records

7.4 Governance and Administration

Governance is kept simple because this is a single-owner academic application. Consistent naming is used for forms and reports (for example Student Profile, Book Inventory, Book Return Report), which makes the data model easier to read. The application owner is responsible for maintaining the forms, correcting data-entry mistakes and keeping sample data separatefrom any real data. Advanced enterprise administration features such as automatedaudit trails and scheduled cloud backups were not required for this small-scale project;manual CSV/spreadsheet export from list reports can be used when record backups areneeded.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The working of the application is demonstrated using one scenario with fictional sample data. The librarian first adds a student through the Student Profile form (Figure 8.1) and maintains the book inventory, which in the sample data holds five titles including “The Great Gatsby”, “Pride and Prejudice” and “1984”. When a book comes back, the librarian opens the Book Return form, enters the return details and selects the Return Condition. For example, return RET-2026-005, made by Eva Martinez on 21-Oct-2026, is recorded with the condition MinorDamage (Figure 8.2).

Finally, the librarian opens the Dashboard to confirm that the new return is counted. In the sample data the dashboard shows five book returns, of which three are Missing and two are MinorDamage, together with three issues, a total fine amount of 0.00 and 870 total copies (Figure 8.3).

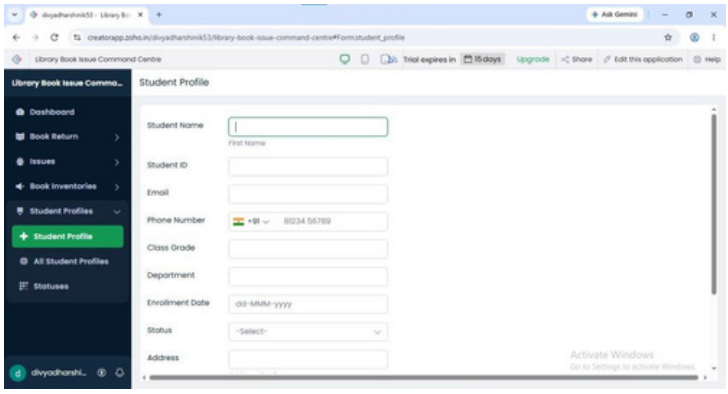


Figure 8.1: Scenario Step 1 – Adding a Student through the Student Profile Form

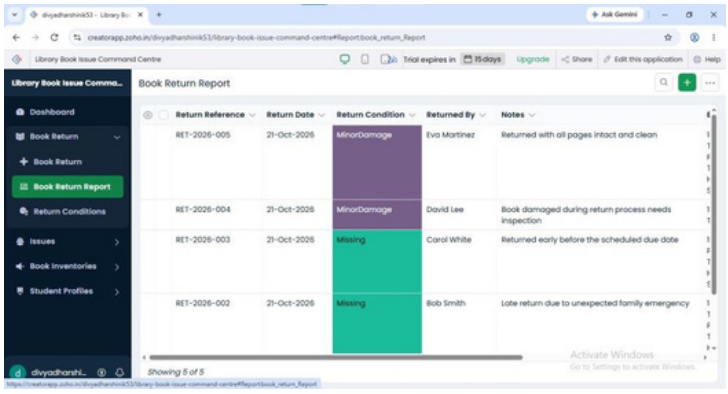


Figure 8.2: Scenario Step 2 – A Book Return Recorded in the Book Return Report

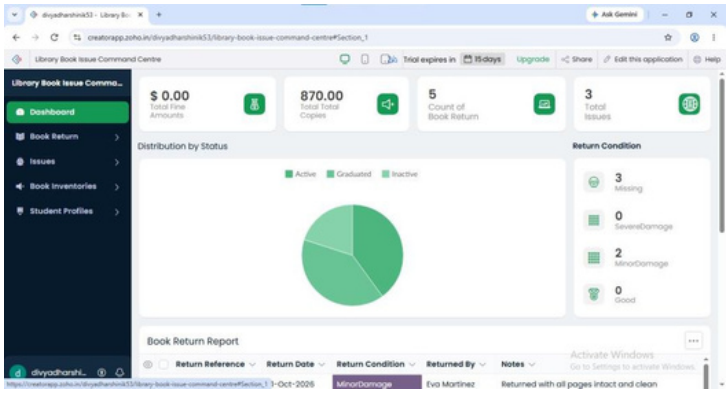


Figure 8.3: Scenario Step 3 – Dashboard Showing the Updated Totals

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	*Implemented Through*	*Outcome / Evidence*
To create a centralised form for student profile information	Section 5.2 – Student Profile form	Form with nine fields, including Status (Figure 5.2)
To maintain a book inventory	Section 5.2 / 5.3 – Book Inventory form and report	Five books listed with copy counts (Figure 5.3)
To record the issue of books	Section 5.2 – Issue form	Total Issues shown on the Dashboard (Figure 5.5)
To record returns with return condition	Section 5.2 / 5.3 – Book Return form and report	Five returns with colour-coded conditions (Figure 5.4)
To classify returns by condition	Return Condition field and Dashboard panel	Counts shown for Missing, SevereDamage, MinorDamage and Good (Figure 5.5; TC-07, TC-09)
To provide reports, a board view and a dashboard	Sections 5.3 and 5.4	Reports and dashboard configured (Figures 5.3 to 5.5)
To demonstrate Zoho Creator low-code features	Chapters 4 and 5	Application built mainly through visual configuration (Section 8.5)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	*Existing System*	*Proposed Application*	*Improvement*
Recording a return	Condition and remarks written in a register or sheet	Entered once in a form with a selectable Return Condition	Consistent, structured records (time not measured)
Manual steps	Separate files for students, books, issues and returns	Related forms in one application	Information kept in one place

Parameter	*Existing System*	*Proposed Application*	*Improvement*
Report generation	Summaries prepared by hand	Reports, board view and dashboard generated from stored records	Consistent, ready-made views
Finding problem returns	Manual search through records	Colour-coded Return Condition and dashboard counts	Damaged and missing returns visible at a glance
Data accessibility	Records available only where the register or file is kept	Available through a browser with the authorised Zoho account	Easier access for the authorised user

8.4 Limitations of the Low-Code Approach

- ***Data dependence:*** the dashboard depends entirely on the accuracy and completeness of the entered records; the Return Condition is chosen by the librarian and is not verified by the system.
- ***No AI or machine learning:*** the application does not use a trained model and does not predict future behaviour.
- ***Platform dependence:*** availability, deployment options and some features depend on the configured Zoho Creator account and edition, and the application cannot easily be moved to another platform.
- ***Customisation limits:*** the layout and charts are limited to the components offered by the Zoho Creator builder; highly customised visualisations would require pro-code extensions.
- ***Access control:*** [custom roles and record-level permissions are not yet configured / describe].
- ***Trial edition:*** the application was built on a trial edition, whose availability is time-limited.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code. Most of the application sits at the no-code end of the spectrum: the four forms, their field types, the list reports, the board view and the dashboard are configured visually without writing code.

Low-code. [Describe any logic expressedas formula fields, workflows or Deluge scripts; if none, say that no custom logic wasneeded.]

Pro-code. No part of the application currently needs pro-code. Moving to pro-code would be justified only if the application required features beyond the platform, such as barcode scanning, integration with an institutional system or a custom interactive visualisation.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Library Book Issue Command Centre demonstrates how a low-code platform can support basic library circulation management. Using Zoho Creator, the project combines structured data entry for students, books, issues and returns with reports, a board view, a return condition classification and a dashboard summary in one application.

The project has strong educational value: it involved data modelling, form design, report and dashboard creation, test planning and reflection on the strengths and limits of low-code development. The dashboard is intentionally simple and transparent; it helps the librarian see the state of the library and identify damaged or missing returns for follow-up.

9.2 Future Enhancements

- Automatic calculation of due dates and fines for late returns (proposed).
- E-mail or scheduled-workflow reminders to students with overdue books (proposed).
- Automatic update of the number of available copies when a book is issued or returned (proposed).
- Secure role-based access for different users, implementing the matrix of Table 4.6 (proposed).
- A student portal for searching the catalogue and viewing personal issue history (proposed).
- Barcode scanning of books and student ID cards (proposed).
- Additional analytics, such as most-borrowed titles and genre-wise circulation (proposed).

9.3 Learning Outcomes and Career Relevance

Building this application strengthened skills in designing a data model, configuring forms with appropriate field types, creating reports, a board view and a dashboard, and planning test cases with positive and negative inputs. It also showed that low-code development still requires careful thinking about data quality, validation and access control.

These skills relate directly to roles such as citizen developer, low-code application developer, solutions consultant and business process automation engineer, where business or educational requirements must be converted quickly into working applications. Knowing when a low-code solution is sufficient — and when pro-code is needed — is also valuable for a future career in software development.

REFERENCES

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	*Zoho Creator Feature*	*Done*	*Where in the Application*	*Report Section*	*Figure / Code No.*
I	Application created & sections/menu organised	Y	Five navigation sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	[Y / N]	Name, Single Line, Email, Phone, Date, Drop Down, Address, Number, … [confirm]	5.2	Tables 4.2–4.5
II	Lookup field (relationship between forms)	[Y / N]	[Issue / Book Return lookups] […]	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	[Y / N]	[…]	4.2, 5.2	—
II	Field validations / layout rules	[Y / N]	List reports used; [filters / grouping / custom actions]	5.2	—
II	List report with filter, grouping, custom action Kanban / Calendar / Timeline / Map report	[Y / N]	Book Inventories Board	5.3	Table 5.1
II	Pivot table / chart report	Y	Distribution by Status pie chart on the Dashboard	5.3	Table 5.1
II	Page / dashboard built with UI Builder	Y	Dashboard page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions) Deluge – functions, fetch, loops, collections	[Y / N]	[…]	5.4	Fig. 5.5
II	Email / in-app notifications	[Y / N]	[…]	5.5	Table 5.2
II	Form workflow	[Y / N]	[…]	5.5	—
II	Scheduled workflow	[Y / N]	[…]	5.5,5.6	—
II	Approval process	[Y / N]	[…]	5.6	Table 5.3
II	Blueprint	[Y / N]	[…]	5.6	—
II	Batch / report workflow Testing & debugging evidence	Y	Testcases and defect log	5.6	—
III	Integration with a Zoho service Connection / REST API call (invokeurl)	[Y / N]	[…]	5.6	—
III	Webhook or Creator REST API	[Y / N]	[…]	5.6	—
III	Users, roles and permission profiles Record-level access criteria / customer portal Zia / AI feature	[Y / N]	[…]	6 5.7	Table 6.1 —

Unit	*Zoho Creator Feature*	*Done*	*Where in the Application*	*Report Section*	*Figure / Code No.*
III	Analytics (charts / Zoho Analytics)	Y	Dashboard totals, status chart and return condition counts (descriptive only)	5.9	Fig. 5.5
IV	Performance optimisation applied & measured	N	Good practices followed; not measured	7.2	Table 7.2
IV	Deployment across environments	[Y / N]	Development environment [confirm]	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	[Y / N]	Zoho sign-in; [others]	7.3	Table 7.3
IV	Governance (audit trail, administration)	[Y / N]	Naming conventions and settings [confirm]	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	[Y / N]	[…]	5.10	—
V	Low-code limitations & spectrum analysis	Y	Limitations and spectrum discussed	8.4, 8.5	Sec. 8.4, 8.5
—	Mobile app / portal access	N	Not tested / not configured	5.11	—

Total features implemented: [__] / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	***Course Outcome Statement***	***Evidenced In***	***Self-Assessed Level***	***Examiner Level***
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	[ ]	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

[Paste the complete Deluge source code of every script used in the application here, with the form and event for each script. If no Deluge script was used, state: “No Deluge scripts were required; the application was built using standard Zoho Creator configuration.”]

```
// Script ID : [DS-01]
// Location : [Form -> Event]

// Purpose   : [Purpose of the script]

// paste code here
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open a web browser, sign in to Zoho Creator with the authorised Zoho account and open the Library Book Issue Command Centre application from the list of applications or via the application URL given in Project at a Glance. The Dashboard is shown first; the other sections (Book Return, Issues, Book Inventories and Student Profiles) are available in the navigation menu. Mobile access has not been tested in the current version.

D.2 Instructions for the Librarian

- Open Student Profiles → Student Profile and add each student with Student Name, Student ID, Email, Phone Number, Class Grade, Department, Enrollment Date, Status and Address; use All Student Profiles to view, edit or search records.
- Open Book Inventories → Book Inventory and add each title with Book Name, ISBN, Author, Publisher, Publication Year, Genre and Total Copies; use All Book Inventories or the Book Inventories Board to view the books.
- Open Issues and record each book issued to a student. [Add the exact steps for your Issue form.]
- When a book is returned, open Book Return → Book Return, enter the return details, select the Return Condition (Good, MinorDamage, SevereDamage or Missing) and submit.
- Open Book Return → Book Return Report to review all returns; returns marked MinorDamage, SevereDamage or Missing are candidates for follow-up.
- Open the Dashboard to view total fine amounts, total copies, the number of returns and issues, the distribution of students by status and the return condition counts.

D.3 Instructions for the Application Administrator

- Sign in with the application owner account and open the application in edit mode (“Edit this application”) to modify forms, fields, reports or the Dashboard.
- After any change, repeat the relevant test cases in Table 6.1 before using the application with real data.

- Keep the application private to authorised users and avoid publishing it publicly, because it contains student information.

APPENDIX E – EVALUATION RUBRIC

(For ExaminerUse Only)

Student Name	***[STUDENT NAME]***	***Register No.***	***[REGISTER NUMBER]***
Project Title	LIBRARY BOOK ISSUE COMMAND CENTRE	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	***Criterion***	***Indicators***	***Max. Marks***	***Marks Awarded***
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		***100***	

Table E.2: Performance Level Descriptors

Level	***% of Max. Marks***	***Descriptor***
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.

Level	*% of Max. Marks*	*Descriptor*
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____

Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For ExaminerUse Only)

Student Name	***[STUDENT NAME]***	***Register No.***	***[REGISTER NUMBER]***
Department / Section	[Department]	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	***Question / Topic Asked***	***Unit***	***Response (E / G / S / NI)***	***Remarks***
1				
2				
3				
4				
5				
6				
7				
8				

E–Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	***Max. Marks***	***Marks Awarded***
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	***100***	

Examiners’ Remarks:

_____

_____

INTERNAL EXAMINER Name: _____ EXTERNAL EXAMINER Name: _____